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Feasibility of Electro-Reduction of Metals in Sulfide-Based Copper Removal Slag for Steelmaking

Zhihao LU,[1,2)]* Xiangwei LIAO,[1,2)] Chenglin ZHAO,[1,2,3)] Deli SHANG,[1,2)] Haiwei WANG[1,2)] and Chengjun LIU[4)]

1) State Key Laboratory of Metal Material for Marine Equipment and Application, Anshan Liaoning,114009 China.
2) Ansteel Iron & Steel Research Institutes, Anshan Liaoning, 114009, China.
3) Now at Ling Yuan Iron & Steel Corporation., LTD., Lingyuan City, Liaoning Province, 122500 China.
4) Key Laboratory for Ecological Metallurgy of Multimetallic Ores (Ministry of Education), School of Metallurgy, Northeastern University, Mailbox 288, Northeastern University, Shenyang Liaoning, 110819 PR China.

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This paper proposed an innovative electrochemical method for copper recovery from spent sulfide slag (Na_2S–FeS–Cu_2S), generated during copper removal from iron-based melts. It aims to reduce the high decopperization agents' consumption caused by low copper distribution ratio during the copper removal process, address the cost issues by recycle of slag and recovery of copper, and expand the application potential of the emerging copper removal method using the sulfide slag systems. The feasibility of electro-recovery of copper is verified by cyclic voltammetry tests and potentiostatic electrolysis experiments. The results show that in the Na_2S–FeS–Cu_2S melt, although copper cannot be reduced alone without reducing iron, rough separation can be achieved by controlling the temperature. The current efficiency of Cu^+ reduction in the molten sulfide is about 9.1% due to electronic conduction in the sulfide melt, which needs further improvement. The slag treated by the innovative electrochemical process can be reused, as the XRD results show that it has a similar composition to the primary slag unused.

KEY WORDS: copper recovery; iron-based melt; electroextraction; molten sulfide electrolyte; recycling of waste.

1. Introduction

Copper is widely used in daily life, especially in electronic products, because of its excellent physical and chemical properties, including high electrical and thermal conductivity, outstanding workability, and corrosion resistance.[1)] However, copper is not always satisfactory, it is one of the most undesirable contaminants in steel because it increases the sensitivity of steel and cannot be removed efficiently and economically.[2)] The widespread use of copper in vehicles, electrical appliances, and equipment has aggravated the copper pollution trend in the steel-making process due to the shorter life circle and the higher recycling efficiency of these steel productions.[3,4)] Copper is stable in the oxidizing atmosphere, the existing refining processes in steel-making cannot remove it. Once copper is introduced to the steel unintentionally, it will accumulate in scrap steel recycling and significantly affect it from the middle of last century.[5)]

A comprehensive comparison highlights the potential of using sulfide slag for copper removal from copper-bearing iron-based melts.[6–12)] However, the distribution ratio of Cu between the sulfide slag and iron-based melt is still too low to apply in industrial. The distribution ratio has been significantly improved with the continuous efforts of researchers, but the copper removal process also poses environmental risks by generating a substantial amount of spent sulfide slag.

Various studies on electrochemical reduction of molten oxides, fluorides, chlorides, *etc.* have emerged in recent decades, highlighting the superiority of electrochemical methods applied to materials and metallurgical disciplines.[13–17)] Some researchers[18–20)] have reported the feasibility of electro-extraction of metals from different all-sulfide melts, the results indicate that components with high electronegativity difference can increase the ionic conductivity of sulfide melts. Based on this, the electrochemical processes in Na_2S–Cu_2S, and Na_2S–Cu_2S–FeS melts are studied in this paper with cyclic voltammetry and potentiostatic electrolysis. The results indicate that the Cu_2S in the spent sulfide slag Na_2S–FeS–Cu_2S can be reduced to metallic copper and elemental sulfur. Therefore, the activity of Cu_2S in the spent sulfide slag, a_{Cu2S}, decreases with the progress of

* Corresponding author: E-mail: lzh_ansteel@163.com

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electrochemical processes, then, the slag can be reused. This process can significantly reduce the amount of waste slag generated when using sulfide slag to treat copper-bearing iron-based melts. The copper and sulfur in this electrochemical process can be recovered as by-products. It should be noted that the traditional smelting of copper by pyrometallurgy emits a large amount of SO_2. Therefore, even if there is a certain amount of waste sulfide slag emitted after copper removal, it can be regarded as a potential copper mineral resource after property treatment, and the electrochemical process proposed in this paper provides the possibility for clean and green production of copper, especially considering the actual situation of copper concentrate shortage in China.[21] The findings in this paper effectively promote the research progress and practical application of copper removal from the iron-based melt using sulfide slag systems.

2. Experimental

In the cyclic voltammetry (CV) tests, a three-electrode system is employed. The electrodes include spectral pure graphite rods (diameter = 3 mm) serving as the working electrodes (WE) and pseudo-reference electrodes (RE), and the bottom of a high-purity graphite crucible as the counter electrodes (CE). The electrode rods are protected by alumina tubes and connected to molybdenum wires for electrical conduction. The alumina crucible with a hole at the bottom is embedded into the graphite crucible, which serves as both the counter electrode (CE) and a protective layer. These components are shown schematically in **Fig. 1**.

The CV tests are conducted in three distinct sulfide melts: the blank group (in a Na_2S melt), group I (in a Na_2S–Cu_2S melt), and group II (in a Na_2S–FeS–Cu_2S melt) at 1 350°C. The compositions of these melts are detailed in **Table 1**. The design of the group II melt is based on previous experimental data from the author's previous work.[22]

The potentiostatic electrolysis experiment is also conducted at 1 350°C in the sulfide melt of group II for 30 minutes with a potential of −0.8 V using the same electrode system. After the test, the electrodes and electrolytes are removed and quenched using compressed argon gas (Ar > 99.9%).

The sulfide melts are synthesized using analytical grade reagents: $Na_2S{\cdot}9H_2O$ (Damao, Tianjin, China), FeS (Damao, Tianjin, China), and Cu_2S (Aladdin, Shanghai, China). Before each experiment, $Na_2S{\cdot}9H_2O$ is dehydrated in a vacuum heating furnace under an argon flow of 1 ml/min at 500°C for over 6 hours. The argon used in these experiments is purified using molecular sieves and activated silica gel to remove residual oxygen and water vapor.

A self-fabricated heating furnace, equipped with $MoSi_2$ heating units and featuring a temperature control accuracy of ±1°C, serves as the heating device. This furnace allows for the sealing of the furnace tube and the injection of argon gas during the experiments. An electrochemical workstation (Metrohm Aut 84847, Switzerland) is utilized to conduct the electrochemical tests detailed above. The potentials obtained from voltammetry are corrected using the measured open circuit potential (OCP).[18,19] The commercial software HSC 6.0 are employed to acquire and calculate the necessary thermodynamic data.

The quenched molten sulfide electrolyte and the products from potentiostatic electrolysis are analyzed using scanning electron microscopy with energy-dispersive spectroscopy (SEM-EDS, Phenom, Prox, Netherlands) and X-ray Diffraction (XRD, Ultima IV, RIGAKU, Japan). The copper and iron content of the product alloy is determined using a method based on the Chinese national standard GB/T 223.18-1994.

For more details about the experimental system, which are not described in this text, please refer to Fig. 1.

3. Results and Discussion

3.1. Thermodynamic Analysis

The standard decomposition voltages, E^0, of Na_2S, FeS, and Cu_2S can be calculated using Formula 1, with thermodynamic data at 1 350°C from HSC Chemistry 6.0, assuming unit activity (a=1) for all components. The calculation results indicate that the thermodynamic sequence of reduction reactions at 1 350°C is FeS/Fe, Cu_2S/Cu, Na_2S/Na, and the respective decomposition voltages are −0.54 V, −0.65 V, and −1.45 V. However, in actual conditions, the decomposition is influenced by the electrolyte's content of each compo-

Table 1. Composition of the sulfide electrolyte melts.

Experimental number	Electrolyte composition
Group I (blank group)	100 mass%Na_2S
Group II	97.5 mass%Na_2S-2.5 mass%Cu_2S
Group III	78.0 mass%Na_2S-2.0 mass%Cu_2S-20.0 mass%FeS

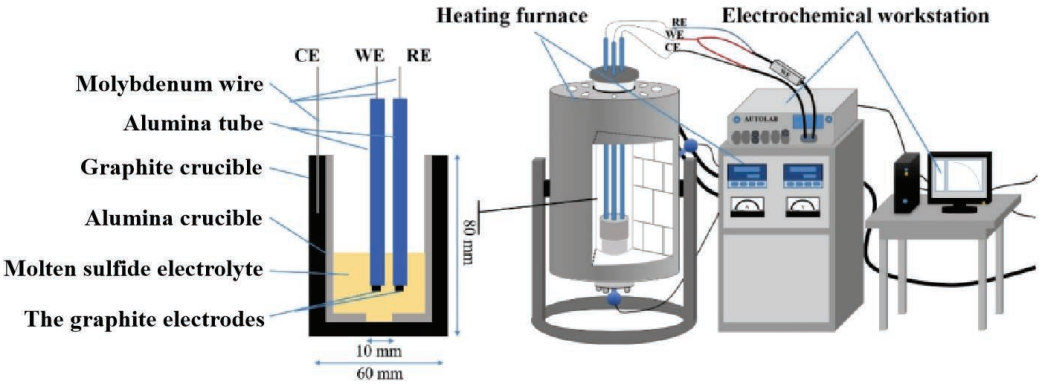


Fig. 1. Schematic diagram of experimental system and electrolysis cell. (Online version in color.)

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nent.[23] The actual decomposition voltages of Cu_2S and FeS in this research are calculated using Formula 2 and 3, taking into account the activity and activity coefficient of FeS and Cu_2S in Na_2S–FeS–Cu_2S melt, based on Wang's works.[10,11] The calculated actual decomposition voltages for FeS and Cu_2S are −0.537 V and −0.649 V respectively, assuming unit activity for reduced metals and sulfur. These finding suggest that copper cannot be electro-reduced independently without the reduction of iron in this system. Therefore, to ensure that electrolytic iron is solid and copper is liquid, the experimental temperature is set to 1 350°C, which allows for the separation of generated copper and iron.

$$E^0 = -\frac{\Delta G^0}{n\mathrm{F}} \quad \text{..... (1)}$$

$$E_{Cu_2S/Cu} = E^0_{Cu_2S/Cu} - \frac{\mathrm{R}T}{n\mathrm{F}} \ln \frac{a^2_{Cu} \cdot a_S}{a_{Cu_2S}} \quad \text{..... (2)}$$

$$E_{FeS/Fe} = E^0_{FeS/Fe} - \frac{\mathrm{R}T}{n\mathrm{F}} \ln \frac{a_{Fe} \cdot a_S}{a_{FeS}} \quad \text{..... (3)}$$

3.2. The CV Tests

The results of cyclic voltammetry are graphically represented in **Fig. 2**. The dotted line depicts the test conducted in molten Na_2S of experimental Group I. As the cathode potential sweeps to approximately −1.5 V (vs. graphite), a reduction peak R1 emerges, which can be attributed to the reduction peak of Na^+. This is due to the sole presence of cation Na^+ in this test group. Notably, the potential of this peak aligns closely with the theoretical decomposition voltage of Na_2S at this temperature, as calculated previously.

In the forward sweep process, an oxidation peak O1 is observed around 1.0 V. This peak corresponds to the oxidation of the anion S^{2-} in the molten sulfide system. Additionally, it aids in determining the electrochemical window for subsequent analysis in the Na_2S–FeS–Cu_2S melt. The Na_2S–Cu_2S melt is then tested within a potential range of 1.5 V to −1.5 V, with the results displayed in the blue line in Fig. 2. A reduction peak R1' is observed at a potential approximately −0.7 V. By comparing it with the results obtained from the pure Na_2S melt, it can be assigned to the reduction peak of Cu^+. No further reduction peaks are observed in this system, indicating that the Na_2S–Cu_2S melt exhibits good electrochemical stability under the conditions detailed in this paper. Finally, the Na_2S–FeS–Cu_2S melt is tested within a potential range of 0.5 V to −1.0 V. The results are presented in the red line in Fig. 2. As the cathodic potential sweeps to approximately −0.4 V, a new reduction peak R1'' appears. The second reduction peak R2'' in this test is found at the potential near to peak R1' in the test of Na_2S–Cu_2S melt. Consequently, the peak R2'' can be attributed to the reduction peak of Cu^+, while the peak R1'' can be assigned to the reduction of Fe^{2+}. The CV test results align with the thermodynamic analysis outcomes detailed previously. The theoretical decomposition voltage of FeS not only surpasses Cu_2S but also the concentration of FeS in the sulfide melt of experimental group III exceeds that of Cu_2S. This disparity results in the preferential reduction of Fe^{2+} prior to the reduction of Cu^+ in the system.

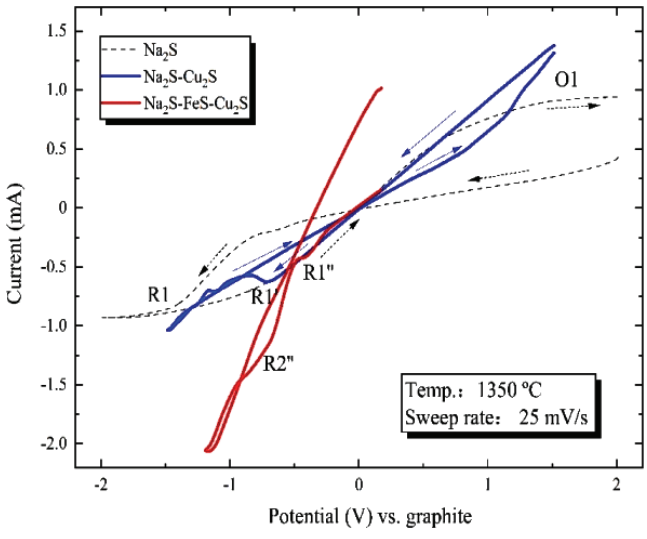


Fig. 2. Cyclic voltammogram in molten sulfide system. (Online version in color.)

Furthermore, the cyclic voltammetry curves obtained in the melt reveal a plateau stage close to zero current during the negative sweep before −1.0 V. Notably, their slope differs significantly from that of a typical molten salt system,[24] indicating that there remains a degree of electronic conductivity within the molten sulfide at 1 350°C.[20] When Cu_2S and FeS are present in the molten sulfide, the slope of the voltammetry curve sharpens, indicating that the electronic conductivity of the molten sulfide electrolyte is enhanced by the addition of Cu_2S with its semiconductor properties and FeS with its electronic conductor properties.

3.3. The Potentiostatic Electrolysis

The sectional image of the graphite cathode and its associated product layer, resulting from potentiostatic electrolysis within the Na_2S–FeS–Cu_2S experimental group, is presented in **Fig. 3**. When observed from a macroscopic perspective, the cathode product adheres to the surface of the graphite cathode in a shell-like formation, exuding a distinctive metallic luster.

The SEM-EDS analysis of a specific point on this product layer, as detailed in **Fig. 4**, reveals that it is primarily composed of Fe and Cu elements. Notably, there is no detection of S, O, or any other elements within this layer, indicating that the product originates from the co-deposition of Fe and Cu. The area enriched with S is attributed to the sulfide electrolyte that has adhered to the cathode.

Furthermore, another type of electrolytic product was observed in the Na_2S–FeS–Cu_2S electrolyte following electrolysis, as seen in **Fig. 5**. The SEM-EDS analysis of these products revealed that they are metallic Cu and Fe. Given that the experimental temperature lies below the melting point of Fe and above that of Cu, the low Cu-containing alloy electrodeposited on the cathode remains solid. Conversely, the high Cu-containing alloys can liquefy within the electrolysis cell, facilitating the pre-separation of copper and iron. Subsequently, this crude copper can be refined into high-purity copper. These findings confirm the feasibility of employing the electrochemical method proposed in this study to recover copper from sulfide slag.

In terms of the anode product, elemental sulfur is

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Fig. 3. Cross section of the graphite cathode. (Online version in color.)

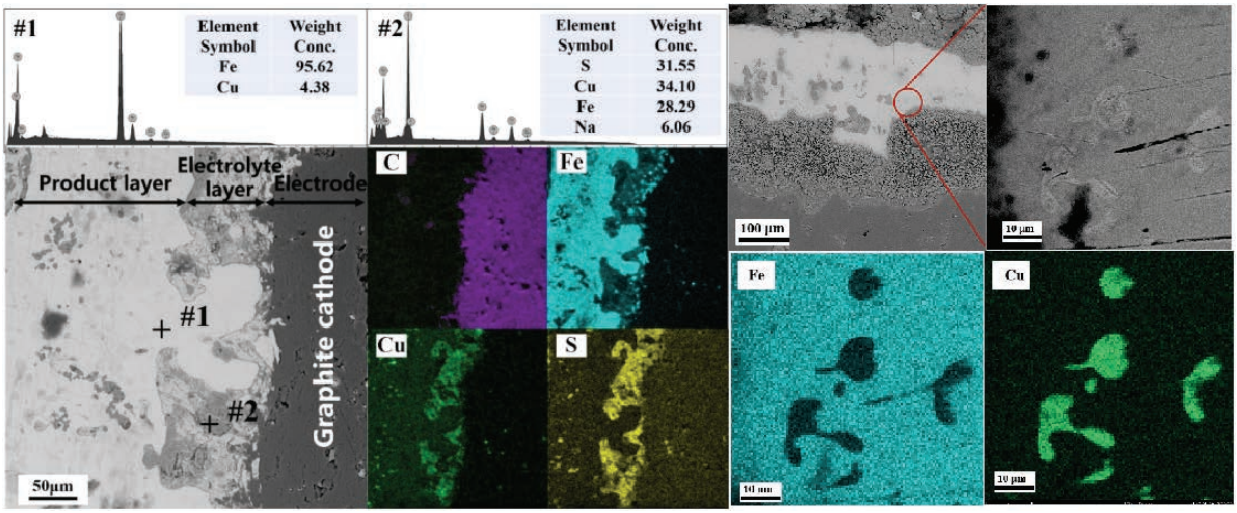


Fig. 4. SEM-EDS analysis of the surface of graphite cathode. (Online version in color.)

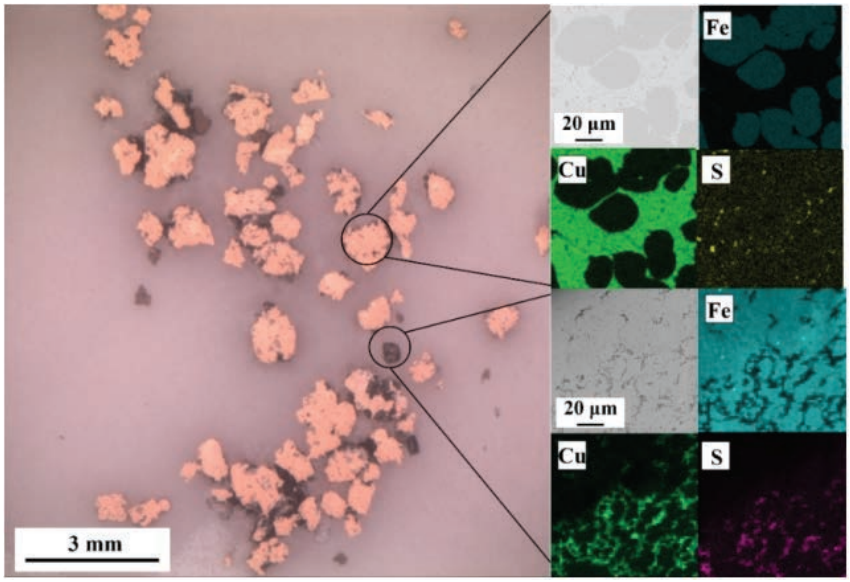


Fig. 5. SEM analysis of the productions in electrolyte. (Online version in color.)

observed accumulating on the anode of the Na_2S–FeS–Cu_2S experimental group, as presented in **Fig. 6**. These findings align with the conclusions derived from CV tests, indicating that the anode reaction during electrolysis in the Na_2S–FeS–Cu_2S melt involves the oxidation of S^{2-} to elemental sulfur. Theoretically, the oxidation reaction on the anode should also include the oxidation of Fe^{2+} to Fe^{3+}. However, there is insufficient evidence to support this reaction from the CV curve. This may be due to the use of graphite crucibles and electrodes in this experiment, which makes the experiment

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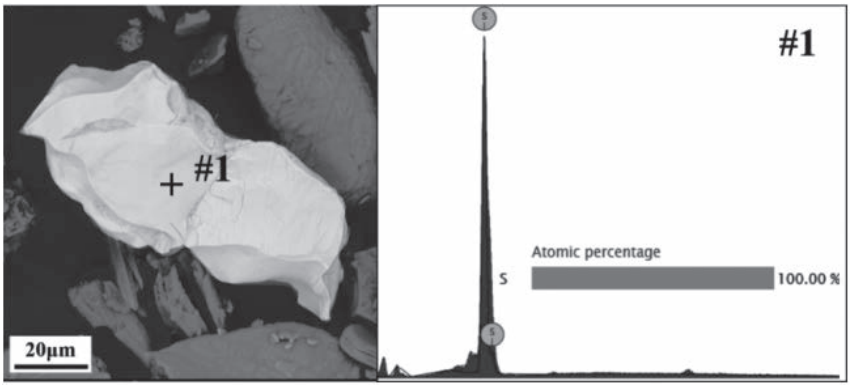


Fig. 6. SEM-EDS analyses of the production on the anode.

conducted in a reducing atmosphere and limits the oxidation of iron ions. However, this issue will be inevitable when the process is promoted industrially. How to limit the oxidation of ferrous iron to avoid unnecessary electron loss and improve the current efficiency of metal reduction is one of the focuses of follow-up research in this study.

Similar anodic processes have been reported in prior studies conducted by Sahu and Sokhanvaran.[18,19] It is worth noting that the temperature in this experiment exceeds the boiling point of elemental sulfur, resulting in most sulfur produced during the electrolysis of metal sulfide escaping as gas. With further research and the development of an effective method for collecting sulfur vapor, it is conceivable that elemental sulfur could also serve as a by-product of this electrolysis process.

In summary, the electrode reactions can be summarized as follows:

Cathode:

$$Cu^{+} + e^{-} \rightarrow Cu \text{ and } Fe^{2+} + 2e^{-} \rightarrow Fe \quad \text{..... (4)}$$

Anode:

$$S^{2-} \rightarrow S + 2e^{-} \quad \text{..... (5)}$$

3.4. The Recovery of the Sulfide Slag

The *i-t* curve for potentiostatic electrolysis within the Na_2S–FeS–Cu_2S melt is presented in **Fig. 7**. A current peak is observed approximately 5 minutes into the electrolysis process. At this juncture, the double electrode layer charging has been completed, leading to a subsequent decrease in current. As the electrolysis progresses, Cu^{+} and Fe^{2+} ions within the molten sulfide gradually undergo reduction. After 10 minutes of electrolysis, the current stabilizes, indicating the emergence of the first current plateau. When the electrolysis reaches approximately 20 minutes, a second current plateau appears, indicating a change in the electrodeposition priority of Fe^{2+} and Cu^{+} due to ion concentration variations during electrolysis.

It is worth noting that the background current for this molten sulfur system stands at approximately 0.18 A, which is higher than that observed in molten salt electrolysis processes. This suggests that the molten sulfide electrolyte of the Na_2S–FeS–Cu_2S melt still exhibits a certain degree of electronic conductivity, potentially impacting the current efficiency of sulfide electroreduction, η_{Cu}. The total charge consumed Q_a throughout the electrolysis process can be

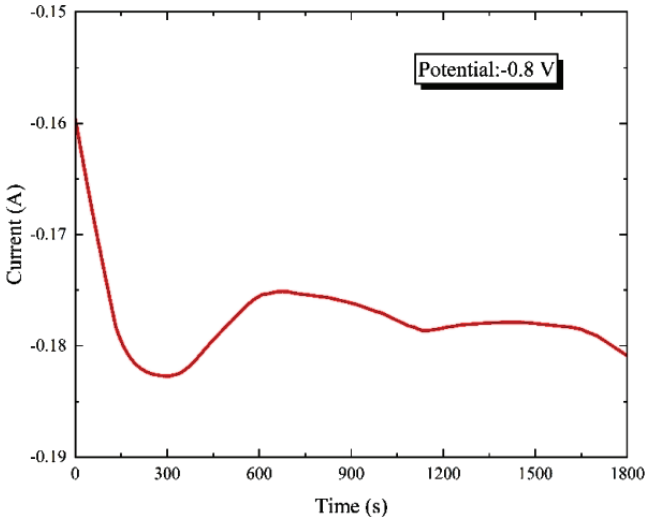


Fig. 7. The *i-t* curve of electrolysis in group II at 1 350°C. (Online version in color.)

determined by integrating the *i-t* curve, amounting to 323 C.

The metal particles collected from the sulfide electrolyte weigh 0.012 g, with the copper content of these particles determined to be 43.5%. The product layer on the cathode surface weighs 0.145 g and contains approximately 9.8% copper. Consequently, the total metallic copper yield from potentiostatic electrolysis within molten Na_2S–FeS–Cu_2S in this study stands at 0.0194 g. The Q_{Cu} can be calculated as 29 C using the Faraday constant of 96 485 C/mol, The current efficiency for copper electroreduction during the electrolysis process can be calculated as 9.1% using Formula 6.[25,26] Where: Q_{Cu} is the charge transfer number for electroreduction of copper during electrolysis, C; Q_a is the total charge transfer number during electrolysis, C; m is the mass of the product obtained by electrolysis, g; M is the molar mass of the product, g/mol. As analyzed before, this numerical value is relatively low due to the electronic conductivity of sulfide slag.

Part of the metallic copper that has been reduced may undergo re-sulfurization within the electrolyte, leading to a potential increase in the overall current efficiency of copper reduction throughout the process. The residual electronic conductivity within the molten sulfide electrolyte and the co-deposition of Fe^{2+} and Cu^{+} during electrolysis remain significant challenges in enhancing the current efficiency of copper electroreduction from the Na_2S–FeS–Cu_2S melt.

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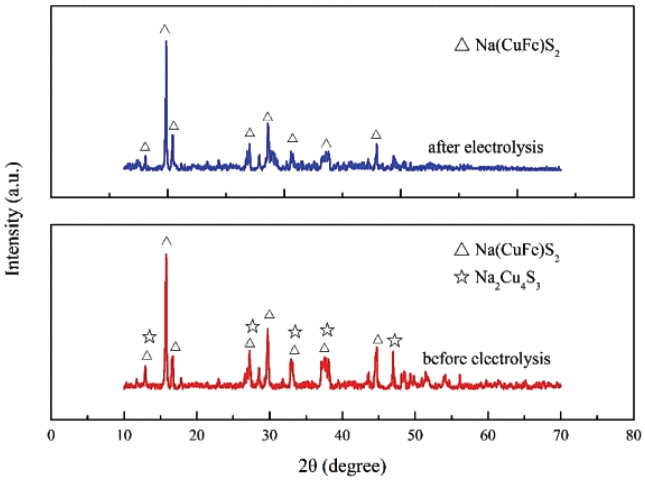


Fig. 8. The XRD results of the sulfide slag. (Online version in color.)

The XRD analysis results for sulfide slag, both before and after electrolysis, are presented in **Fig. 8**. These findings indicate that the main component of the sulfide slag after electrolysis remains primarily sulfide, sharing similarities with the slag prior to electrolysis. This suggests that the sulfide slag can indeed be reused as decopperization material. The method proposed in this study offers a means of slag reutilization. Notably, the characteristic peak for the primary copper-containing phase, $Na_2Cu_4S_3$, within the slag appears to be diminished following electrolysis, aligning with the reduction in copper content within the sulfide melt throughout the electrolysis process.

$$\eta_{Cu} = \frac{Q_{Cu}}{Q_a} = \frac{n\mathrm{F}\dfrac{m_{Cu}}{M_{Cu}}}{Q_a} \times 100\% \quad \cdots\cdots\cdots\cdots (6)$$

4. Conclusion

This paper provides evidence of the feasibility of electrolyzing Cu_2S within the sulfide decopperization slag, Na_2S–FeS–Cu_2S, through CV tests and potentiostatic electrolysis. The key findings under the specified conditions in this paper are as follows.

Following copper removal from a copper-bearing iron-based melt, the Na_2S–FeS–Cu_2S slag can be served as the electrolyte for Cu_2S and FeS. The cathode product is metallic Cu and Fe, the copper is liquid due to its lower melting point, while the iron is solid and adheres to the graphite cathode. The anode product is elemental sulfur. It is feasible to recover copper from the sulfide decopperization slag, enabling its reuse in the copper removal process within refining in steel production. The current efficiency of copper electro-reduction from the sulfide slag stands at 9.1%. Future studies should aim to improve the efficiency of copper recovery from the sulfide decopperization slag and facilitate its sustainable reuse in industries.

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